

RAY OPTICS

* LIGHT

- Light is a form of energy, which enables us to visualise the things, but light itself is not visible.
- Speed of light does not depend on speed of source & observer.
- frequency & sense of colour does not depend on medium.
- Intensity, wavelength, speed of light depends on medium.

$$uv = c = \text{constant}$$

$$u_1 v_1 = u_2 v_2$$

$$v = \lambda f$$

$$u \propto \frac{1}{\lambda}$$

- [VIBGYOR] $\rightarrow$ 400nm to 700nm
- Monochromatic light $\rightarrow$ light of single λ ?
- White light $\rightarrow$ combination of all wavelength.
- At crossing point direction of ray do not change.
- formation of Image does not depend upon size of mirror but visibility of image depends on size of mirror.

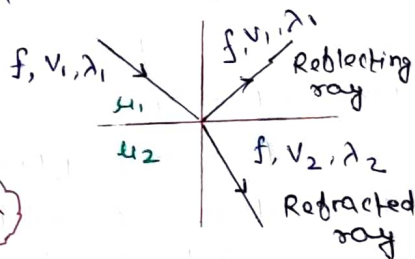
PLANE MIRROR

Effect of Reflection & Refraction

f = Medium independent, only depend on source.

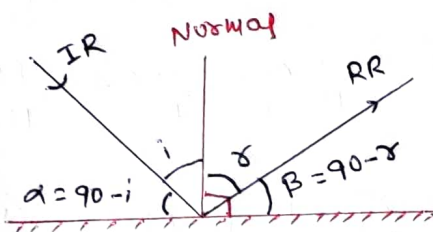
$$v = \frac{c}{u} = \lambda f$$

$$v \propto \frac{1}{u} \quad \lambda \propto \frac{1}{u}$$



Laws of Reflection

- $\angle i = \angle r \rightarrow$ Measured from normal.
- Normal, incident ray (IR) & reflected ray (RR) all are lie in same plane.

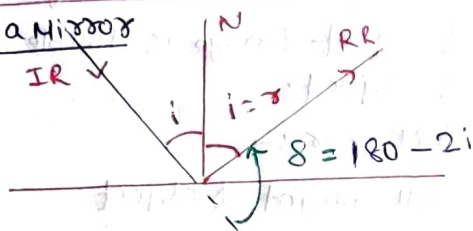


$\hookrightarrow$ Angle b/w IR & RR is always $2i$?

$$\hookrightarrow \alpha = \beta$$

Deviation by plane mirror

(a) By a Mirror



(b) By two inclined mirror

$$\delta = 360 - 2\theta$$

$\rightarrow$ does not depend on Angle of incident.

θ = Angle b/w Mirror.

* NOTE:-

- For given incident ray; if the plane mirror is rotated through an angle θ , in the plane of incidence, the reflected ray turn through an angle 2θ in same sense.
- For given mirror, if the incident ray is rotated through an angle θ ; then reflected ray turn through angle θ in opposite sense.
- If mirror is rotated in its own plane then no effect on reflected ray.

Image formation by plane mirror

Object $\rightarrow$ A line $\perp$ to mirror draw karo 3rd line $\overleftrightarrow{O I}$ virtual image hogi.

- Virtual image
- Erect
- Real object $\rightarrow$ virtual image $\rightarrow$ diverging ray
- Virtual object $\rightarrow$ Real image $\rightarrow$ converging ray
- Same size
- laterally inverted

Image formation of extended object

$\hookrightarrow$ Height of Image = Height of object

$$\text{Magnification (M)} = \frac{\text{size of Image}}{\text{size of object}}$$

$$M_T = \frac{H_I}{H_O}$$

$\rightarrow$ If $M_T = +ve \rightarrow$ Erect

$\rightarrow$ If $M_T = -ve \rightarrow$ Inverted

Unitless
dimensionless

If $M_T = +2 \rightarrow$ Erect Image

Size of object Image is double of size of object.

Height of Mirror :-

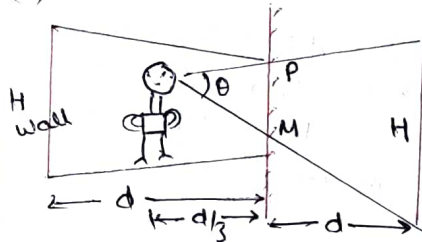
(A) To see full height of object

$$H_{\text{mirror}} = \frac{H_{\text{object}}}{2}$$

(B) To see full wall behind object

$$H_{\text{mirror}} = \frac{H_{\text{wall}}}{3} \quad (\text{observer at centre})$$

(C)



$$\tan \theta = \frac{H/4}{d/3} = \frac{H}{d+d/3}$$

$$PM = \frac{H}{4}$$

Clock system

$$H_I : M_I : S_I = (11 - H_c) : (59 - M_c) : (60 - S_c)$$

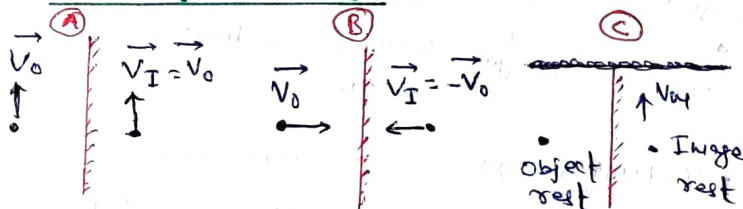
$\Rightarrow$ If only hr hand & Minute hand given

$$H_I : M_I = (11 - H_c) : (60 - M_c)$$

$\Rightarrow$ If only hr hand is given, then

$$H_I = (12 - h) \text{ hr}$$

Velocity of Image



$$\vec{V}_I = 2\vec{V}_M - \vec{V}_O$$

$$\vec{V}_{IO} = \vec{V}_I - \vec{V}_O$$

If $V_O = \text{rest} \rightarrow \vec{V}_{IO} = 2\vec{V}_M$

Image due to combination of two plane mirror :-

$n = \text{no. of Images obtained}$

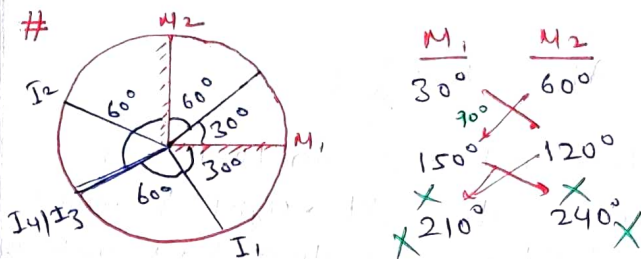
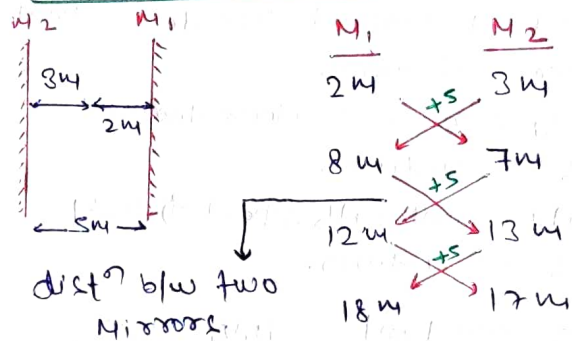
(i) If $m = \frac{360^\circ}{\theta}$ is even

$$n = m - 1$$

(ii) If $m = \frac{360^\circ}{\theta}$ is odd

$$n = m - 1 \rightarrow \text{symm.} \quad n = m \rightarrow \text{unsymm.}$$

Infinite Image when two plane mirror is parallel :-



Spherical Mirror

Concave Mirror

Converging

$$f = -ve$$

$$U = -ve [RO]$$

$$V = -ve [RI]$$

Convex Mirror

Diverging

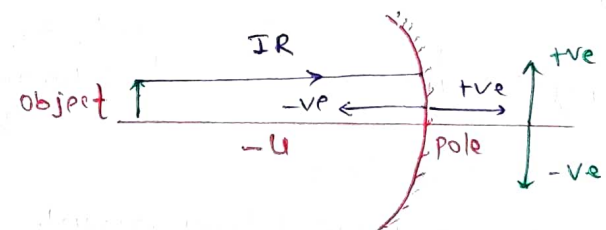
$$f = +ve$$

$$U = -ve [RO]$$

$$V = +ve [VI]$$

* NOTE :-

The distance measured in a direction of IR is +ve.



Mirror Equation

$$f = \frac{R}{2} \quad \frac{1}{V} + \frac{1}{U} = \frac{1}{f}$$

$$M_T = \frac{H_I}{H_O} = -\frac{V}{U} = \frac{f}{f-U} = \frac{f-V}{f}$$

$$M_L = \frac{dv}{du} = -m \frac{2}{T} \rightarrow \text{only valid for small object}$$

$M_T \rightarrow$ Ies to principle-axis.

$M_T = +ve \rightarrow$ Erect / virtual

$M_T = -ve \rightarrow$ Inverted / Real

Image formation by concave mirror

Object	Image
At Infinity	At focus, Real, diminished $M = -ve$, Inverted
blw C & ∞	blw C & F , Real, small $M < -1$, Inverted
At C	At C , Real, small $M = -1$, inverted
blw C & f	blw ∞ & C , Real, large $M \geq -1$, inverted
At f	at ∞ , Real, very large $M \gg -1$, Inverted
blw f & pole	behind mirror (virtual), large, $M = +ve$, Erect

Image formation by convex mirror

Object	Image
at ∞	At focus, virtual $M = +ve$, Erect
blw ∞ & pole	blw pole & focus

Velocity of Image in a case of curved Mirror:-

① object is moving $\perp$ to principle axis

$$V_I = M_T V_o$$

② object is moving $\parallel$ to principle axis

$$V_I = M_T^2 V_o$$

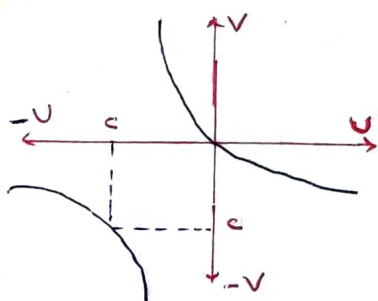
NOTE:-

object is at $\infty \rightarrow M_T = 1$, $V_I = V_o$

object blw C & $f \rightarrow M_T > 1$, $V_I > V_o$

object blw C & $\infty \rightarrow M_T < 1$, $V_I < V_o$

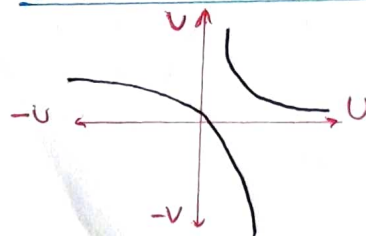
Graph blw U & V for concave mirror



* for Real object
image always form
blw pole & focus
for convex mirror

* virtual Image $\rightarrow$
concave
mirror.

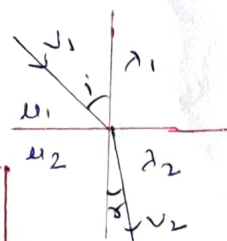
for convex mirror



Refraction:-

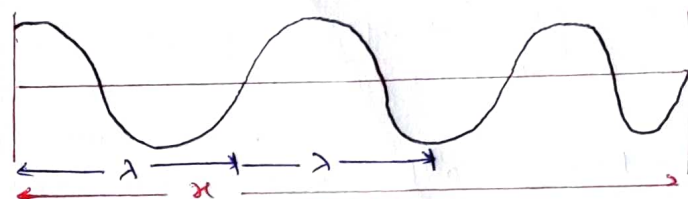
(1) $R \rightarrow D (\delta = i - r)$

(2) $D \rightarrow R (\delta = r - i)$



$$\mu_2 = \frac{v_2}{\lambda_2} = \frac{\mu_1}{\lambda_1} = \frac{v_1}{\lambda_1} = \frac{v_2}{\lambda_2}$$

$\Rightarrow$ No. of wave in a given length 'x'



$$n\lambda = x$$

$$n = \frac{x}{\lambda}$$

Image due to Refraction at a plane surface:-

The object is in denser medium & is seen from rarer medium normally through plane surface, the apparent depth, $= d(1 - \frac{1}{\mu})$

$$\frac{d_{app}}{\mu_{R|obj}} = \frac{d_{real}}{\mu_{R|obj}} \rightarrow d_{shift} = d_{real} - d_{app}$$

Refraction through a glass slab

① Normal shift $\rightarrow t(1 - \frac{1}{\mu})$

② Lateral shift $\rightarrow (R \rightarrow D \rightarrow R)$

$$d = \frac{t}{\cos r} \sin(i - r) \rightarrow \text{If } i \rightarrow \text{small}$$

$$d = ti [1 - \frac{1}{\mu}]$$

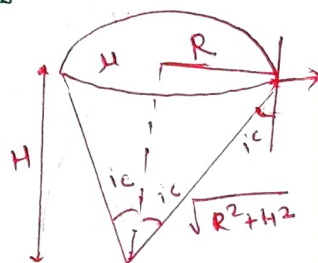
Total internal Reflection [$i > i_c$]

$\rightarrow$ Light must travel from $D \rightarrow R$

$$\sin i_c = \frac{\mu_2}{\mu_1} = \frac{v_1}{v_2} = \frac{\lambda_1}{\lambda_2}$$

Radius of visibility

$$R = \frac{H}{\sqrt{u^2 - 1}}$$



Optical fibre → Based on TIR

$u_1 > u_2$
core cladding

$$\sin i_c = \sqrt{u_1^2 - u_2^2}$$

for TIR → $\sin i \leq \sqrt{u_1^2 - u_2^2}$

Refraction at spherical surfaces

$$\frac{u_{RR}}{v} - \frac{u_{LR}}{u} = \frac{u_{RR} - u_{LR}}{R}$$

→ Use with proper sign convention.

Concave → $R = -ve$

Convex → $R = +ve$

* Magnification: -

$$m_T = \frac{H_I}{H_O} = \frac{u_{LR} \cdot v}{u_{RR} \cdot u}$$

$$m_L = \frac{dv}{du} = \frac{u_{LR} \cdot v^2}{u_{RR} \cdot u^2}$$

Lens Maker Equation

$$\frac{1}{v} - \frac{1}{u} = \left(\frac{u_L}{u_M} - 1 \right) \left[\frac{1}{R_1} - \frac{1}{R_2} \right] \quad (u_M = 1)$$

Lens Equation

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \quad \frac{1}{f} = \left(\frac{u_L}{u_M} - 1 \right) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

Convex → $f = +ve$

Concave → $f = -ve$

* Biconvex lens

$$f = \frac{R}{2(u-1)}$$

* Plano-convex

$$f = \frac{R}{(u-1)}$$

→ Focal length of lens depend upon medium.

$u_M \rightarrow$ Medium, $u_L \rightarrow$ lens

$u_M = u_L \rightarrow$ lens behave as simple glass

$u_M > u_L \rightarrow$ diverging lens (opposite)

$u_L > u_M \rightarrow$ Nature of lens remain same.

→ Glass lens in air medium, Biconvex lens

$$f = R$$

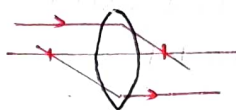
Magnification

$$m_T = \frac{H_I}{H_O} = \frac{v}{u} = \frac{f}{f+u}$$

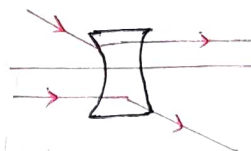
$$m_L = m_T^2 = \frac{dv}{du}$$

→ for small object

Converging lens



Diverging lens



Mirror	Lens
$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$	$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$
$m_T = \frac{H_I}{H_O} = -\frac{v}{u}$	$m_T = \frac{H_I}{H_O} = \left(\frac{v}{u} \right)$
$m_T = \frac{f}{f-u}$	$m_T = \frac{f}{f+u}$
$m_T = \left(\frac{f-v}{f} \right)$	$m_T = \left(\frac{f-v}{f} \right)$
$f = \frac{R}{2} \rightarrow$ does not depend on Medium	depends on Med.

Image formation by Convex lens

Convex lens $\xrightarrow{\text{con (diverging)}}$ Concave mirror

Object	Image
At ∞	at focus, small, real, inverted, $m = -ve$
blw $2f$ & ∞	blw f & $2f$, small, real, inverted, $m = -ve$
blw $2f$ & f	blw $2f$ & ∞ , large, real, inverted, $m = -ve \uparrow$
At $2f$	at $2f$, same size, real, inverted, $m = -1$
At f	at ∞ , very large, real, inverted, $m = -ve \uparrow \uparrow$
Blw f & pole	Same side, large, virtual, erect, $m = +ve$

Image formation by Concave lens

Object	Image
At ∞	at focus, virtual, erect, $m = +ve$
blw ∞ & pole	blw pole & focus

concave lens $\Rightarrow$ concave mirror
(diverging)

Power

Len

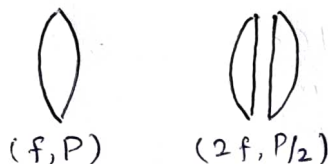
$$P = \frac{1}{f(\text{cm})} = \frac{100}{f(\text{cm})}$$

SI UNIT $\rightarrow$ Dioptre (D)

Mirror

$$P = -\frac{1}{f(\text{cm})} = -\frac{100}{f(\text{cm})}$$

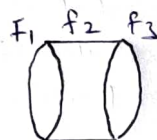
Cutting of lenses



Combination of lens

(a) When lens are in contact:-

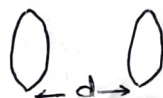
$$P = P_1 + P_2 + P_3$$



$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3}$$

(b) When lens are separated by a distance 'd'

$$P = P_1 + P_2 - d P_1 P_2$$



$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}$$

Displacement Method:- (To find focal length)

$$f = \frac{D^2 - x^2}{4D}$$

$$H_0 = \sqrt{I_1 I_2}$$

$$|f| = \frac{m_1 - m_2}{x}$$

$$m_1 = \frac{v}{u} = \frac{I_1}{H_0}$$

$$m_2 = \frac{v}{u} = \frac{I_2}{H_0}$$

D $\rightarrow$ distⁿ b/w object & screen
x $\rightarrow$ shift in the position of lens

Combination of a lens & a mirror (Silvering of Lens):-

$$\frac{1}{|f_{\text{net}}|} = \frac{2}{|f_L|} + \frac{1}{|f_M|}$$

Convex lens $\rightarrow$ concave mirror
Silvered [f = -ve]

Concave lens $\rightarrow$ convex mirror
Silvered [f = +ve]



$$f = \frac{-R}{2(\mu - 1)}$$



$$f = \frac{-R}{2\mu}$$



$$f = \frac{-R}{2(2\mu - 1)}$$



$$f = \frac{R}{2(\mu - 1)}$$



$$f = \frac{R}{2\mu}$$



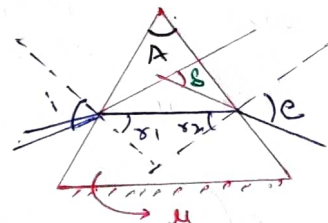
$$f = \frac{R}{2(2\mu - 1)}$$

Prism

A = Angle of prism
= refracting Angle

$$A = r_1 + r_2$$

$$\delta = i + e - A$$

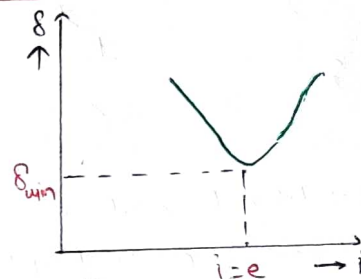


At minimum deviation

$$i = e$$

$$\delta_{\text{min}} = 2i - A$$

$$\mu = \frac{\sin\left(\frac{A + \delta_{\text{min}}}{2}\right)}{\sin(A/2)}$$



for thin prism, A is small

$$\delta_{\text{min}} = (\mu - 1)A$$

$\Rightarrow$ If thin placed in medium of refractive index μ_m .

$$\delta = A\left(\frac{\mu_p}{\mu_m} - 1\right)$$

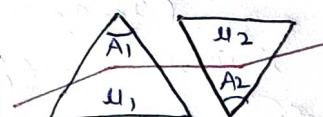


$$\delta = A(\mu_p - 1) \rightarrow \text{In air}$$

Condition for no deviation:-

$$\delta_1 + \delta_2 = 0$$

$$A_1(\mu_1 - 1) = -A_2(\mu_2 - 1)$$



Dispersion

- In vacuum speed of all colour is same equal to $3 \times 10^8 \text{ m/s}$.
- But in medium, diff. colour travel with different speed, because 'u' is diff for diff colour.

$$u = A + \frac{B}{\lambda^2}$$

VIRGYOR $\rightarrow$
 $\lambda \uparrow \mu \downarrow$

$$\mu \propto \frac{1}{\lambda}$$

Mean deviation (δ_{mean})

$$\delta = A(\mu - 1) \quad \theta = A(\mu_v - \mu_r)$$

$$\delta_{\text{mean}} = \frac{\delta_v + \delta_r}{2}$$

$$\delta_{\text{mean}} = A[\mu_{\text{avg}} - 1]$$

$$\delta_{\text{mean}} = A[\mu_y - 1]$$

$$\mu_{\text{avg}} = \frac{\mu_r + \mu_v}{2} = \mu_y$$

Dispersive power:- (ω)

$$\omega = \frac{\theta}{\delta_{\text{mean}}} = A \left[\frac{\mu_v - \mu_r}{\mu_y - 1} \right]$$

Angular Dispersion (θ)

$$\theta = \delta_v - \delta_r$$

$$\delta_{\text{mean}} = A[\mu_y - 1] = A \left[\frac{\mu_r + \mu_v}{2} - 1 \right]$$

Dispersion without deviation

$$\delta_{\text{net}} = \delta_1 + \delta_2 = 0$$

$$A_1(\mu_y^1 - 1) = -A_2[\mu_y^2 - 1]$$

Optical Instruments

* Visual Angle :- Angle from horizontal with which an observer see an object

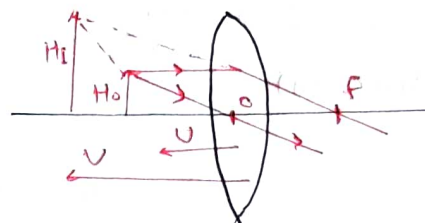
$$\theta_0 = \frac{H_0}{D}$$

$$D = 25 \text{ cm}$$

→ dist of distinct vision.

Simple Microscope

- Use as a magnifier
- Convex lens → (small focal length)
- Converging lens of short focal length
- Erect, magnified & virtual image
- Image is at infinity
 - Relaxed eye position
 - Comfortable eye
 - far point → $M_{\text{min}} = \frac{D}{f_e}$
- Image is at 25 cm
 - Strained eye
 - Near point $M = 1 + \frac{D}{f_e}$



$$\text{Magnifying power (M)} = \frac{\theta}{\theta_0}$$

$$M = \frac{D}{u}$$

$$M = D \left[\frac{1}{f} + \frac{1}{v} \right]$$

Compound Microscope

Objective lens

- Near the object
- forms real, inverted, magnified image of obj.
- Small focal length

The eyepiece

- large focal length
- Use as a simple microscope
- forms large virtual image

$$M_o = M_o M_e = \frac{V_o}{U_o} \frac{D}{U_e} \quad M = \frac{D V_o}{U_o} \left[\frac{1}{f_e} + \frac{1}{V_e} \right]$$

$$\textcircled{1} v = D \text{ (near)}$$

$$M = \frac{V_o}{U_o} \left(1 + \frac{D}{f_e} \right)$$

$$\textcircled{2} v = \infty \text{ (far)}$$

$$M = \frac{V_o D}{U_o f_e} = \frac{L D}{f_o f_e}$$

Length of compound microscope

$$L = V_o + U_e = V_o \quad \text{near} \quad L = V_o + f_e \quad \text{far}$$

Astronomical telescope

$$M_o = \frac{f_o}{U_e} = \frac{f_o}{f_e} \left(1 + \frac{f_e}{V_e} \right)$$

$$\textcircled{1} v = D \text{ (near)}$$

$$M = \frac{f_o}{f_e} \left(1 + \frac{f_e}{D} \right)$$

$$L = f_o + U_e$$

$$\textcircled{2} v = \infty \text{ (far)}$$

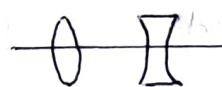
$$M = \frac{f_o}{f_e}$$

$$L = f_o + f_e$$

Galileo Telescope

$$L = f_o + f_e$$

Rest same



@cyanotes ♥